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Energy Balance

$$\frac{\partial}{\partial t}\sum w_i + \cdot \sum \mathcal{S}_{w_i} + j \cdot \mathcal{E} = 0 \tag{1}$$

Abbreviations:

w_i = engery densities (electric, magnetic, gravitational, etc. ...), $i \in \mathbb{N}$

$\mathcal{S}_{w_i}$ = Poyinting-Vectros $\hat{=}$ energy transport densities (transfers) by fields (electric, magnetic, gravitational, etc. ...) to corresponding energy densities w_i

$j \cdot \mathcal{E}$ = energy gain provided by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

Mechanical

...

Electromagnectic energy transfer

charge conservation:

$$\frac{\partial}{\partial t}\varrho + \nabla \cdot j = 0 \tag{2}$$

$\varrho = \sigma\delta$ = charge density in interior

σ = charge density at border (surface)

δ = Dirac delta-function

$j = \sigma\left(\mathcal{E} + v \times \mathcal{B}\right) = \nabla \times \mathcal{H} - \frac{\partial}{\partial t}\mathcal{D} \hat{=}$ elecectric current density

$j \cdot \mathcal{E} \hat{=}$ energy gain by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{S} = \mathcal{E} \times \mathcal{H} \hat{=}$ energy transport density (Intensity = $|\bar{\mathcal{S}}|$)

Energy conservation for (large extended) conductive medium:

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$$\int \left(\frac{\partial}{\partial t}\left(w_e + w_m\right) + \ \cdot \mathcal{S} + j \cdot \mathcal{E}\right)\, dr = \frac{d}{dt}W_e + \frac{d}{dt}W_m + D_S + \int \left(j \cdot \mathcal{E}\right)\, dr \overset{\text{low frequency}}{=} \frac{1}{2}\frac{d}{dt}\left(LI^2\right) + \frac{1}{2}\frac{d}{dt}\left(\frac{Q^2}{C}\right) \tag{3}$$

Abreviations:

$$\Delta\cdot := \text{ div grad} = \text{Laplacian}$$

$$J = \text{"ignition of inhomogenity,cause"} \quad j = \text{Current density} \quad \varrho = \text{Charge density}$$

$$V = \text{Potentialdifference} \quad I = \text{Current} \quad R = \text{Resistance}$$

$$L = \text{Inductance} \qquad C = \text{Capacitance} \qquad Q = \text{charge (ion)} \qquad \Psi = LI = \text{mag. Flux}$$

$$\mathcal{S} = \mathcal{E} \times \mathcal{H} = \text{Poynting-Vector, radiation intensity}$$

$$D_S = \int \left(\ \cdot \mathcal{S}\right)\, dr = \int \text{div}(\mathcal{E}\times\mathcal{H})\, dr = \text{dissipation engergy lost from radiation} \approx 0 \text{ for low frequencies } \omega$$

$$d = \sqrt{\frac{1}{\mu\sigma}\frac{2}{\omega}} = \text{skin depth (surface) of conductor} \qquad r = \frac{d}{\sqrt{2i}}\rho = \frac{d}{1+i}\rho = \text{radius (extension) of conductor}$$

$$\rho = kr = \text{"radius of curvature } k\text{"}$$

$$\mu = \text{permeability of conductor} \qquad \sigma = \text{conductivity of conductor (dielectricum)}$$

Thermal

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Ralativistc

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